

Necessity Entrepreneurship: transplant introduction preview

1 Introduction

Why do some workers become entrepreneurs only when paid employment is unavailable, and what does that imply for policy? This question lies at the center of debates over labor markets, small business, and social insurance, yet necessity entrepreneurship is difficult to identify in observed data. The object of interest is inherently counterfactual. A worker is a necessity entrepreneur if they enter business ownership when a viable employment relationship is unavailable, but would prefer wage work if such a relationship were available. Standard startup data do not reveal that counterfactual. The same observed entry may reflect entrepreneurial comparative advantage, temporary self-employment during job search, insurance against labor-market risk, or a response to job loss. These cases are unlikely to have the same implications for firm scale, growth potential, or welfare.

Empirical work has approached this identification problem in two main ways. One uses stated motives. The Global Entrepreneurship Monitor (GEM) provides nationally representative survey evidence on why individuals start businesses and shows that necessity-type motives are quantitatively important even in the United States. In weighted U.S. APS microdata for 2019–2021, the share of early-stage entrepreneurs endorsing the motive "to earn a living because jobs are scarce" is 40.9 percent in 2019, 49.6 percent in 2020, and 45.7 percent in 2021. At the same time, 68.3 percent, 64.7 percent, and 73.4 percent, respectively, endorse the motive of building great wealth or very high income. The coexistence of these responses is informative rather than contradictory: it shows that necessity-type considerations are common, but also that necessity and opportunity signals need not be mutually exclusive within the same entrepreneurial spell. The most recent U.S. GEM report points in the same direction, documenting a sharp rise in job-scarcity motives between 2021 and 2023 (Brush et al., 2024). Cross-country evidence is equally suggestive. The GEM 2015/16 global report shows that 78 percent of entrepreneurs in innovation-driven economies are opportunity-motivated, and that the ratio of improvement-driven opportunity to necessity entrepreneurship is much higher in richer economies than in factor-driven ones (Kelley et al., 2016). Related evidence on entrepreneurial aspirations likewise suggests that the motives behind entry vary systematically with development (Chivers, 2017).

A second empirical strategy classifies entrepreneurs by observed labor-market status. Fairlie and Fossen (2019) define opportunity entrepreneurship as entry from employment and necessity entrepreneurship as entry from non-employment, and show that the two components have different cyclical properties and different growth orientation. Fairlie (2013) and Fossen (2021) similarly find that business creation is more countercyclical on margins associated with unemployment-driven and unincorporated entry. These classifications are tractable and informative, but they are not equivalent to the counterfactual object of interest. A worker who expects separation may start a business while still employed and appear opportunity-driven in the data, while a non-employed worker may enter temporarily while continuing to search and appear necessity-driven

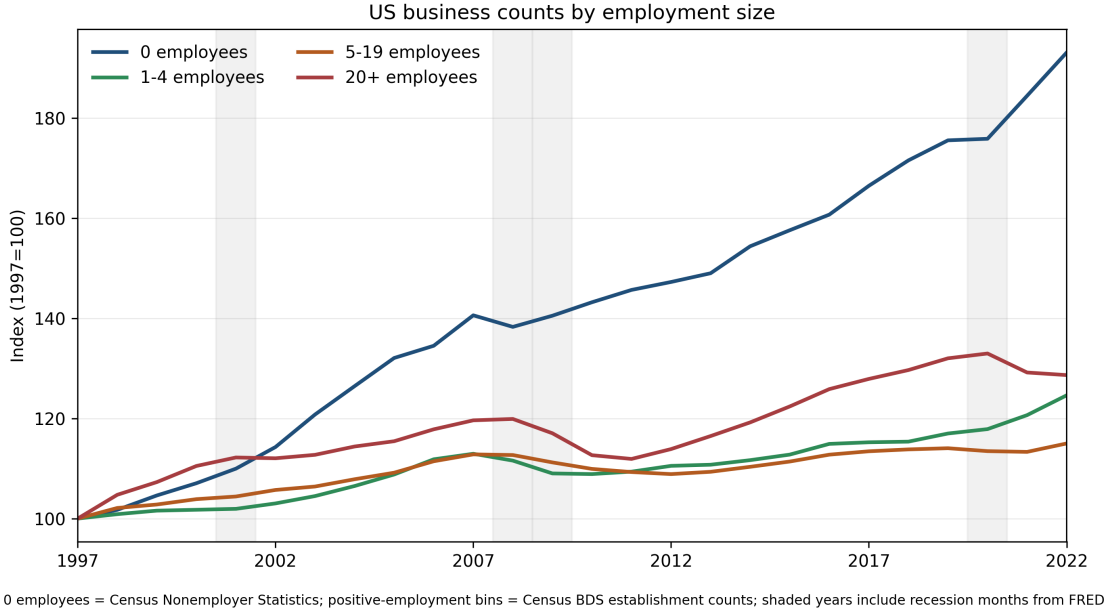


Figure 1: U.S. business counts by employment size, indexed to 2007 = 100. The zero-employee margin is comparatively resilient in the Great Recession, while employer establishments contract.

even when their long-run entrepreneurial comparative advantage is high.

This paper takes survey-based and status-based classifications seriously, but treats them as proxies rather than definitions. We instead adopt a structural, state-contingent definition. An opportunity entrepreneur is an agent who chooses business ownership whether they enter the period employed or non-employed. A necessity entrepreneur is an agent who chooses business ownership only when they lack an employment relationship. The distinction maps directly into occupational choice, does not require taking survey answers literally, and does not equate necessity mechanically with entry from non-employment. We embed that distinction in a quantitative occupational-choice framework with entrepreneurial ability, human capital, search frictions, and unemployment insurance so that the model speaks jointly to entry, entrant type, and firm scale.

Business-cycle evidence is useful in this setting because downturns compress outside options, but recession patterns are best viewed as motivation rather than definition. Our empirical side evidence is consistent with that view. In U.S. business counts by employment size, the Great Recession is associated with relative resilience at the zero-employee margin and contraction across employer establishments. Between 2007 and 2010, zero-employee establishments rise by about 1.9 percent, while establishments with 1–4 employees fall by 3.6 percent, establishments with 5–19 employees fall by 2.6 percent, and establishments with 20 or more employees fall by 5.8 percent. Aggregate counts move little because growth at the zero-employee margin offsets declines among employer establishments. The natural interpretation is not that recessions simply create more firms, but that weaker outside options shift activity toward low-scale business ownership. Figure 1 illustrates this compositional pattern directly.

The stock evidence is complemented by flow evidence. Using Census Business Formation Statistics, total business applications decline over the 2007–2010 window, while high-propensity applications decline much more sharply. In our side evidence, total applications fall by about 5.8 percent and high-propensity applications fall by about 22.1 percent, producing a sizable decline

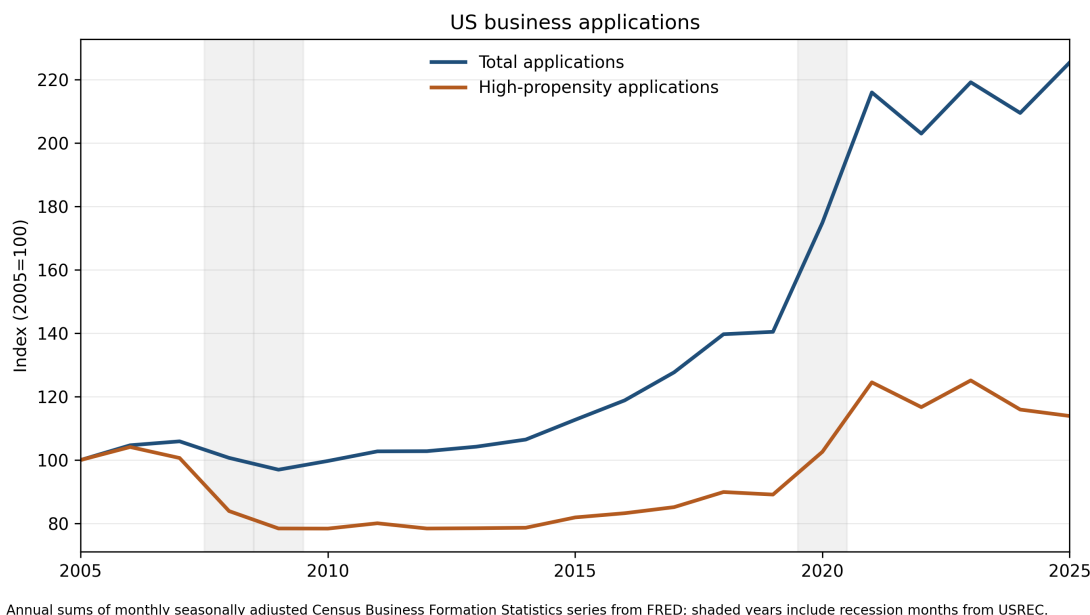


Figure 2: U.S. business applications, indexed to 2007 = 100. Total applications decline over the recession window, and high-propensity applications fall much more sharply.

in the high-propensity share of all applications. This pattern aligns with the broader startup-dynamics literature. Sedlacek and Sterk (2017) show that startup cohorts born in weak aggregate conditions are persistently smaller and have lower growth potential. Haltiwanger et al. (2013) show that startups and young firms are disproportionately important for job creation, while Fort et al. (2013) and Zarutskie and Yang (2017) document that young firms were especially weak during the Great Recession. Recession conditions may therefore preserve or increase low-scale entry while simultaneously weakening the formation of employer firms with stronger growth prospects. Figure 2 places this point on the flow margin.

That distinction is important because low-scale business ownership is not the same thing as employer-firm creation. Fairlie and Miranda (2016) show that most nonemployer startups do not hire, and that transitions from nonemployer to employer status are concentrated among more growth-oriented businesses. Hurst and Pugsley (2011) likewise show that the typical small business is not organized around innovation or expansion. A rise in zero-employee activity should therefore not be read mechanically as evidence of greater productive dynamism. More generally, the recession evidence suggests that the relevant empirical comparison is across margins: own-account or zero-employee business activity, employer-oriented startup formation, and entrant scale or growth potential.

Against this background, we develop a heterogeneous-agent occupational-choice model with entrepreneurial ability, human capital, search frictions, separation risk, and unemployment insurance. Employment relationships are valuable but risky. Unemployment insurance partly insures the downside of job loss, but it also changes the return to waiting for wage work rather than creating a business immediately. Entrepreneurship is therefore an endogenous response to labor-market risk, incomplete insurance, and heterogeneity in productive ability. This approach is closely related to the occupational-choice and entrepreneurial heterogeneity literatures associated with Buera (2009), Cagetti and De Nardi (2006), and Poschke (2013), but it adds an explicit

distinction between entrepreneurship chosen with and without a viable employment relationship. The model therefore allows downturns and policy changes to affect not only the entrepreneurship rate, but also the composition of entrepreneurs and the scale of the firms they operate.

The paper makes three contributions. First, it provides a structural definition of necessity entrepreneurship that maps directly into state-contingent occupational choice. Second, it embeds that definition in a quantitative framework with heterogeneous education, entrepreneurial ability, search frictions, and social insurance, allowing the model to speak jointly to entry, firm scale, and labor-market conditions. Third, it interprets necessity entrepreneurship as a margin of misallocation: weak outside options can push some households into business ownership even when their productivity is better deployed in paid employment. In that sense, the key issue is not only how much entrepreneurship an economy generates, but what type of entrepreneurship it generates.

This perspective matters for policy. Changes in unemployment insurance need not move all entrepreneurship measures in the same direction. A more generous safety net may reduce low-scale necessity entry without reducing productive dynamism if it improves worker-firm matching and lowers business creation driven by weak outside options rather than genuine entrepreneurial comparative advantage. For the same reason, recession-style comparative statics should be judged by whether they reproduce the observed compositional shift toward zero-employee activity and away from employer-oriented formation, rather than by whether they move a single aggregate entrepreneurship statistic in isolation.

The remainder of the paper proceeds as follows. Section 2 discusses the related literature. Section 3 presents the model environment. Section 4 characterizes household behavior and equilibrium. Section 5 describes the calibration. Section 6 presents the baseline results, Section 7 studies the policy counterfactuals, and Section 8 concludes.

References

- Candida G. Brush, Andrew C. Corbett, Caroline Daniels, Donna J. Kelley, Mahdi Majbouri, and Jeffrey Shay. Global entrepreneurship monitor 2023–2024 united states report: National entrepreneurship assessment for the united states of america. Technical report, Babson College, 2024.
- Francisco J. Buera. A dynamic model of entrepreneurship with borrowing constraints: Theory and evidence. *Annals of Finance*, 5(3-4):443–464, 2009.
- Marco Cagetti and Mariacristina De Nardi. Entrepreneurship, frictions, and wealth. *Journal of Political Economy*, 114(5):835–870, 2006.
- David Chivers. Success, survive or escape? aspirations and poverty traps. *Journal of Economic Behavior & Organization*, 143:116–132, 2017.
- Robert W. Fairlie. Entrepreneurship, economic conditions, and the great recession. *Journal of Economics & Management Strategy*, 22(2):207–231, 2013.
- Robert W. Fairlie and Frank M. Fossen. Defining opportunity versus necessity entrepreneurship: Two components of business creation. Working Paper 26377, National Bureau of Economic Research, 2019.

- Robert W. Fairlie and Javier Miranda. Taking the leap: The determinants of entrepreneurs hiring their first employee. Working Paper 22428, National Bureau of Economic Research, 2016.
- Teresa C. Fort, John Haltiwanger, Ron S. Jarmin, and Javier Miranda. How firms respond to business cycles: The role of firm age and firm size. *IMF Economic Review*, 61(3):520–559, 2013.
- Frank M. Fossen. Entrepreneurship over the business cycle in the united states: A decomposition. *Small Business Economics*, 57(2):943–963, 2021.
- John C. Haltiwanger, Ron S. Jarmin, and Javier Miranda. Who creates jobs? small versus large versus young. *Review of Economics and Statistics*, 95(2):347–361, 2013.
- Erik Hurst and Benjamin Wild Pugsley. What do small businesses do? *Brookings Papers on Economic Activity*, 43(2):73–142, 2011.
- Donna Kelley, Slavica Singer, and Mike Herrington. Global entrepreneurship monitor 2015/16 global report. Technical report, Global Entrepreneurship Research Association, 2016.
- Markus Poschke. Who becomes an entrepreneur? labor market prospects and occupational choice. *Journal of Economic Dynamics and Control*, 37(3):693–710, 2013.
- Petr Sedlacek and Vincent Sterk. The growth potential of startups over the business cycle. *American Economic Review*, 107(10):3182–3210, 2017.
- Rebecca Zarutskie and Tiantian Yang. How did young firms fare during the great recession? evidence from the kauffman firm survey. In *Measuring Entrepreneurial Businesses: Current Knowledge and Challenges*, pages 179–211. University of Chicago Press, 2017.